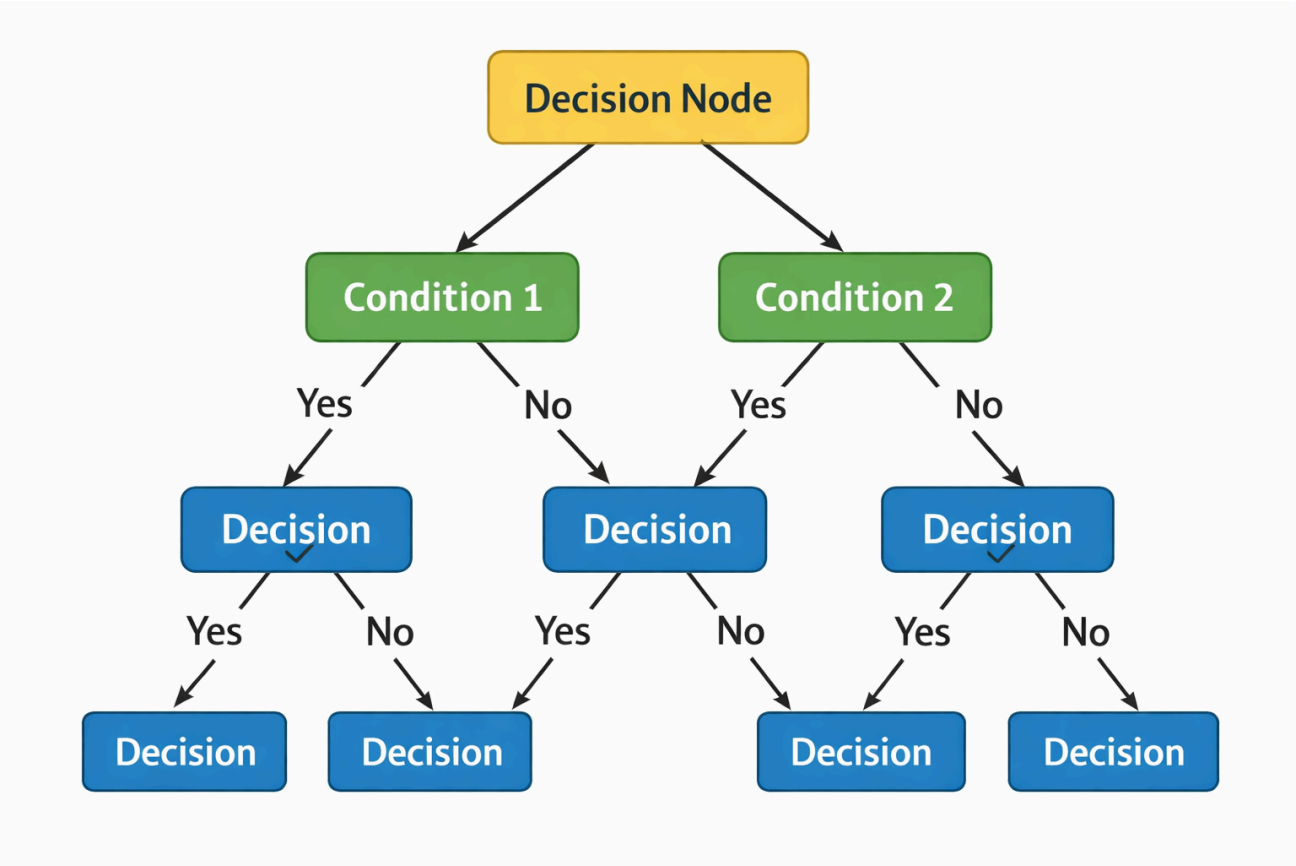
Documentation-Decision Tree

A **Decision Tree** is a supervised learning algorithm used for **classification and regression** that models decisions through a tree-like structure. It recursively partitions the feature space into smaller regions using feature-based splitting rules, aiming to minimize prediction error.



1 Problem Formulation

Input Space:
Feature vector X ∈ R^d (for numerical features) and/or categorical feature space.

Output Space:

- Classification: Y ∈ {1, 2, …, K}
- Regression: Y ∈ R

Data Distribution:
Assume samples (X, Y) are drawn independently from an unknown joint distribution P(X, Y).

Learning Objective (Expected Risk Minimization):
Find a function f(X) that minimizes the expected prediction error:

Expected Risk = E[L(Y, f(X))]

Since the true distribution is unknown, we minimize empirical risk over training data.

2 Model Specification

Hypothesis Function:
A decision tree partitions the feature space into disjoint regions R1, R2, ..., Rm.

Prediction function:

For classification:
$f(X)$ = majority class in region Rj

For regression:
$f(X)$ = mean of Y values in region Rj

Parameter Space:

- Tree structure (splitting features and thresholds)
- Split thresholds
- Leaf node predictions

Structural Assumptions:

- Recursive binary partitioning
- Axis-aligned splits (feature-wise splits)
- Non-linear decision boundaries
- No assumption of linear separability

3 Loss Function

For Classification

Common impurity measures:

Gini Impurity:
$\text{Gini} = 1 - \Sigma (pk^2)$

Entropy:
$\text{Entropy} = - \Sigma (pk \log pk)$

Where pk is the proportion of class k in a node.

For Regression

Mean Squared Error (MSE):
$\text{MSE} = (1/n) \Sigma (yi - y_mean)^2$

Why this loss? (Statistical Reasoning)

- Gini and Entropy measure class purity.

- MSE measures variance reduction in regression.
- Tree chooses splits that maximize impurity reduction.

Convexity Properties:

- Local split optimization is greedy.
- Global tree optimization is non-convex.
- No guarantee of globally optimal tree.

4 Objective Function

Empirical Risk Expression:

Minimize total impurity across all leaf nodes:

Total Loss = Σ (weighted impurity of each leaf)

Regularized Formulation (if pruning used):

Cost Complexity Pruning:

Objective = Empirical Loss + α × (number of leaf nodes)

Where α controls tree complexity.

5 Optimization Method

Closed-form or Iterative?
Greedy recursive partitioning (iterative splitting).

Gradient Expression:
Not gradient-based. Uses impurity reduction criterion:

Information Gain = Parent Impurity − Weighted Child Impurity

Convergence Guarantees:

- Converges in finite steps.
- No guarantee of global optimum.

Computational Complexity:
Training complexity ≈ O(n d log n)
Where:
n = number of samples
d = number of features

6 Statistical Interpretation

MLE / MAP Connection:

- Classification tree approximates Maximum Likelihood under multinomial distribution.
- Regression tree corresponds to Gaussian noise assumption with MSE loss.

Noise Model Assumption:

- Regression: Gaussian noise assumption.
- Classification: Multinomial distribution assumption.

Probabilistic Meaning of Outputs:

- Class probabilities estimated as class proportions in leaf node.
- Regression output is conditional mean estimate.

7 Regularization & Generalization

Bias–Variance Tradeoff:

- Shallow tree → High bias, low variance
- Deep tree → Low bias, high variance

Overfitting Behavior:

- Fully grown trees overfit easily.
- Pruning reduces overfitting.

Capacity Control Mechanism:

- Maximum depth
- Minimum samples per leaf
- Minimum samples for split
- Pruning parameter (α)

8 Theoretical Properties

Convexity / Global Optimality:

- Non-convex optimization problem.
- Greedy algorithm does not guarantee global minimum.

Consistency:

- Under certain conditions, trees are consistent estimators.
- Large trees with proper pruning improve consistency.

Stability Considerations:

- High variance model.
- Small changes in data can significantly change structure.
- Ensemble methods (Random Forest) improve stability.

9 Computational Complexity

Training Time Complexity:
O(n d log n) average case

Inference Time Complexity:
O(tree depth) per sample

Memory Complexity:
O(number of nodes)

10 Limitations

When it fails:

- Small datasets with high variance
- Highly noisy data
- Smooth decision boundaries (piecewise constant limitation)

Assumption Violations:

- Axis-aligned splits may not capture oblique boundaries efficiently

Sensitivity Issues:

- Sensitive to small data perturbations
- Can overfit without pruning
- Biased toward features with many levels (in classification)